



National Textile University

Department of Computer Science

Subject:

Operating System

Submitted to:

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Submitted by:

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Reg number:

23-NTU-CS-1126

Lab no:

14

Semester:

5th

Project Structure:

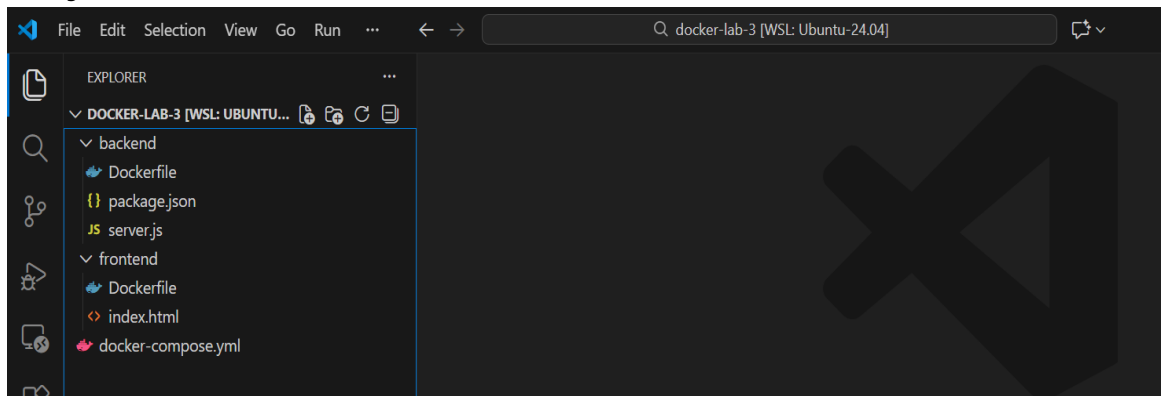


Image creation Frontend:

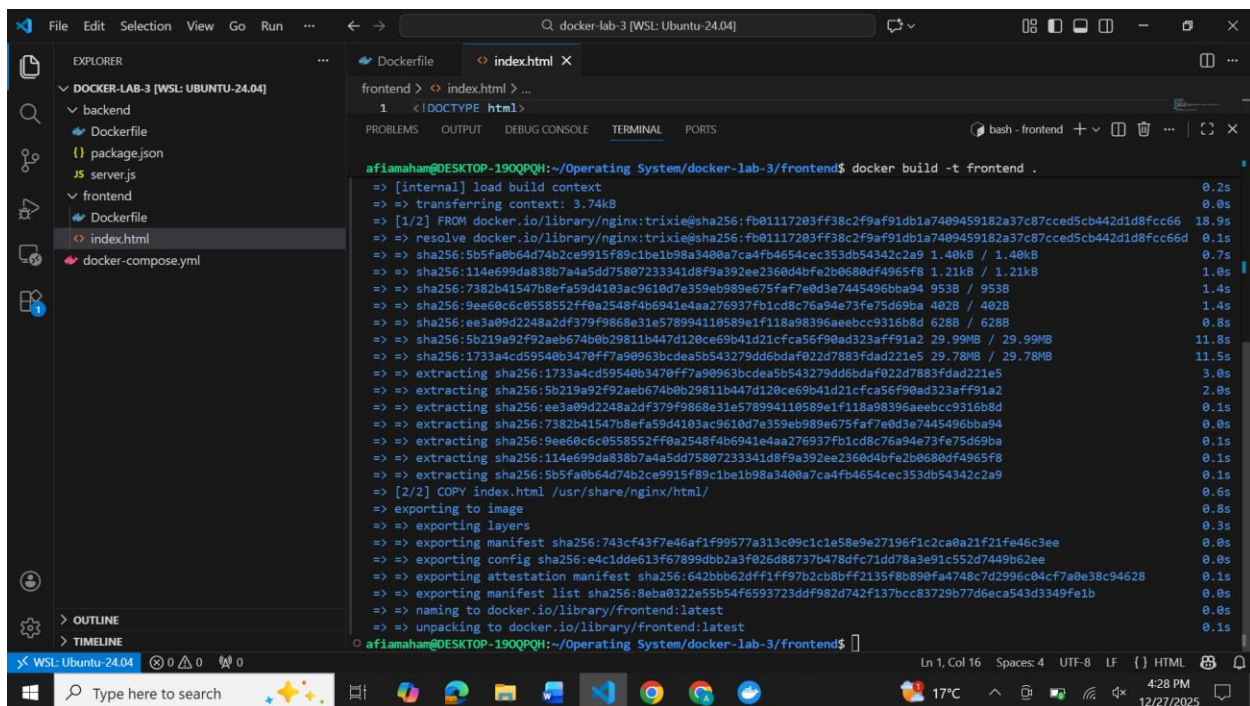
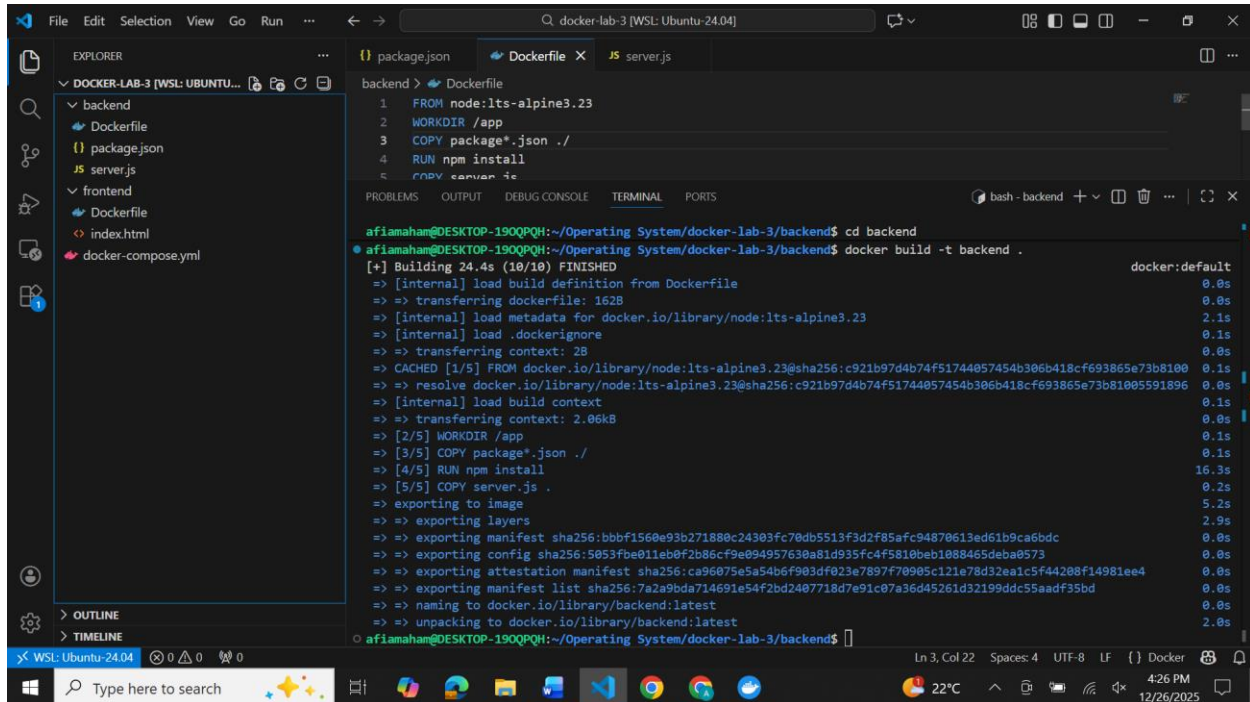


Image creation Backend:

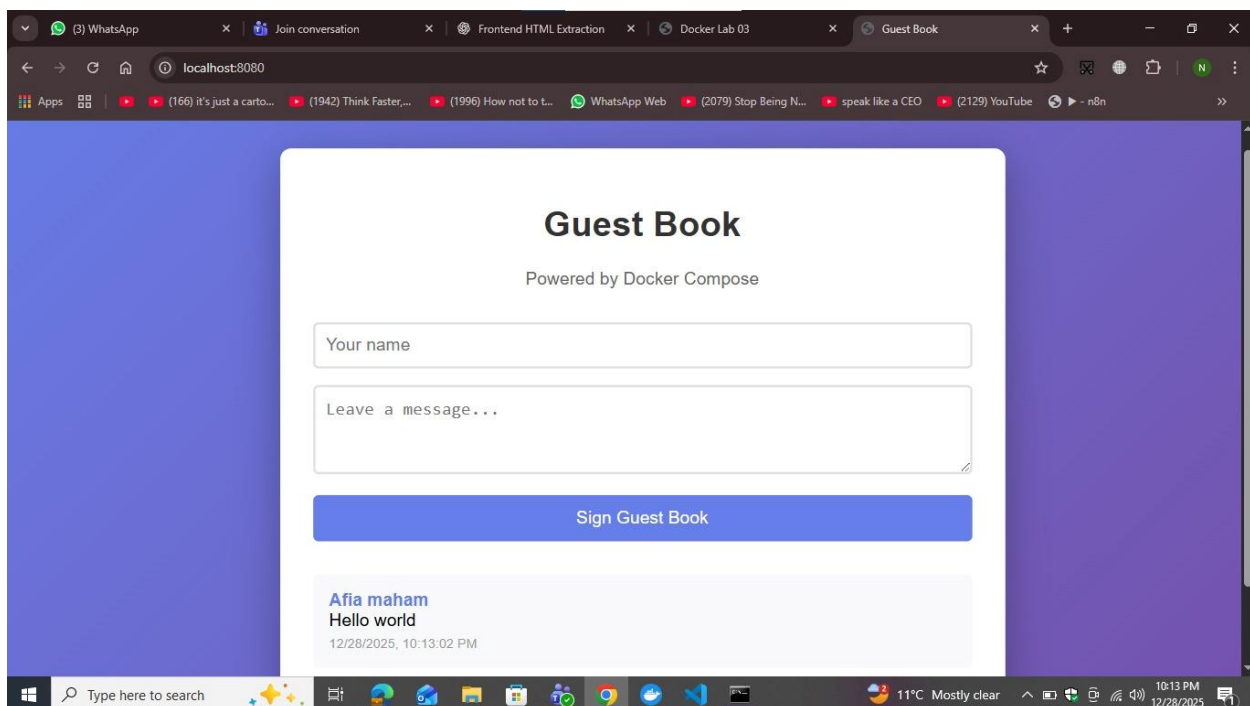


The screenshot shows a VS Code editor with a file explorer on the left showing a project structure for 'DOCKER-LAB-3'. The main editor displays a 'Dockerfile' with the following content:

```
1 FROM node:lts-alpine3.23
2 WORKDIR /app
3 COPY package*.json ./
4 RUN npm install
5 COPY server.js .
```

The terminal at the bottom shows the command `docker build -t backend .` being executed, resulting in a successful build of the 'backend' image. The output includes details about the build process, such as the image being built from 'node:lts-alpine3.23' and the final image being exported to the Docker registry.

Webpage:



Docker-compose content:

services:

MongoDB Database

mongodb:

image: mongo:7-jammy

container_name: guestbook-db

networks:

- <enter_network>

volumes:

- mongo_data:/data/db

Backend API

api:

build: ./backend

image: <image_name>

container_name: guestbook-api

environment:

- MONGO_URL=mongodb://mongodb:27017

ports:

- "3000:3000"

depends_on:

- mongodb

networks:

- <enter_network>

Frontend

web:

build: ./frontend

image: <image_name>

container_name: guestbook-web

ports:

- "8080:80"

depends_on:

- api

networks:

- <enter_network>

networks:

<enter_network>:

volumes:

mongo_data:

Dockerfile Frontend:

FROM nginx:trixie

COPY index.html /usr/share/nginx/html/

EXPOSE 80

Dockerfile Backend:

FROM node:lts-alpine3.23

WORKDIR /app

COPY package*.json ./

RUN npm install

COPY server.js .

EXPOSE 3000

CMD ["npm", "start"]